

8. [Maximum mark: 6]

(a) Given that $w \in \mathbb{C}$, prove that $ww^* = |w|^2$. [2]

Two complex numbers z and w satisfy the following equations:

$$5w^* = (1 - 2i)z^2$$

$$zw = 10 + 10i.$$

(b) Given that $|w| = 2\sqrt{5}$, find z expressing your answer in the form $a + bi$, where $a, b \in \mathbb{Z}$. [4]

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